The Swap Rule

Ganesha, the Janus tetrahedra of squashed stackings, and the border at true close packing



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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).

Abstract. Stack layers of balls, cubic or hexagonal in any order, and share the space between the balls' holes by weight: each hole gets a piece. Paper #57 showed that at true close packing an octahedral piece is two halves, each borrowed from a pure stacking, while a tetrahedral piece is borrowed whole from its nearest layer. Here the layers are squashed or stretched. We prove that for every layer spacing from half the close-packed squash upwards, every hole piece, tetrahedral ones included, is a top half from the pure stacking of the layer above and a bottom half from the pure stacking of the layer below. Two consequences follow. The **Swap Rule**: the two mixed pieces always add up to the two pure ones. And a sharp **border**: the tetrahedral pieces are whole exactly when the layers are at least as far apart as in true close packing; squash them even slightly and each mixed tetrahedral piece gets its head from one side and its body from the other. DICTO calls these pieces **Ganesha**. Below half the squash the halves stop being pure, yet the Swap Rule for volumes still holds, down to a quarter of the squash exactly. The rule was found by DICTO's idea of looking for simple rules in pictures of computed values; a map of eleven spacings shows where each rule holds.

1. The idea

DICTO's idea (27 October 2026): the rule of halves of #57 (DICTO DOI 10.5281/zenodo.23647898) was spotted in a picture of computed values; other such pictures may hide other simple rules (halves, thirds, ...). A rule finder was built to look for them (Splash Rules, study 2026-10-27-splash-rules). Its first run, on the #57 walk with the layer spacing changed, found a rule nobody had looked for: when the layers are squashed, the mixed tetrahedral pieces stop being whole, yet their volumes still add up exactly. DICTO named the rule the Swap Rule and the new pieces Ganesha, after the god whose head was swapped.

2. Set-up

Balls of spacing 1 lie in triangular layers at positions A, B, C, stacked by a word with no letter twice in a row; the layer spacing is d, with $d^2 = s \cdot ⅔$ (s = 1 is true close packing). Each gap between a layer X and the next layer Y holds three holes per cell: an octahedral hole over the third position at mid-gap, a tetrahedral hole over X at ¾ of the gap and one under Y at ¼. As in #57 the octahedral holes get weight 0 and the tetrahedral holes weight λ, and space is divided by power distance (a Laguerre or power diagram); each hole's share is its *piece*. A layer is **c** (cubic) if its two neighbours differ, **h** (hexagonal) if they are equal; the pure stackings are ABC (all c) and AB (all h).

3. Main theorem

> **Theorem.** Let $s \geq \tfrac{1}{2}$. Put $\lambda_-(c) = -\tfrac{1}{4} - s/8$, $\lambda_-(h) = -\tfrac{1}{3} - s/8$ and $U(s) = \min(\tfrac{1}{3} + s/24, 3s/8)$. In every close-packed stacking, periodic or not, take any hole piece of a gap whose lower layer has type t and upper layer type t′, at any λ with $\max(\lambda_-(t), \lambda_-(t')) \leq \lambda \leq U(s)$. Cut it by the horizontal mid-plane of its gap. The part below is congruent to the lower half of the piece of the same kind in the pure stacking of type t, and the part above to the upper half of that piece in the pure stacking of type t′. This holds for all three kinds of hole piece.

The weight range contains the whole walk of the cubic stacking for every $s \geq \tfrac{1}{2}$, and the whole walk of every stacking for $s \geq 1$.

Corollary 1 (the Swap Rule). Write V(t|t′) for the volume of a piece over a t layer and under a t′ layer. Then, for all three kinds of piece,
V(c|h) + V(h|c) = V(c|c) + V(h|h).
Proof: V(t|t′) = L(t) + R(t′) with L and R the pure lower and upper half volumes; both sides equal L(c) + L(h) + R(c) + R(h).

Corollary 2 (the border at true close packing). For $s \geq 1$ the far half of a tetrahedral piece is the same over a c layer as over an h layer, so the piece is borrowed whole from its nearest layer, as in #57. For every s with $\tfrac{1}{2} \leq s < 1$ there is a weight in the cubic walk where the far half over a c layer is a proper part of the far half over an h layer, with strictly smaller volume: the mixed tetrahedral pieces are then Janus c/h pieces in their own right. These are the **Ganesha** pieces.

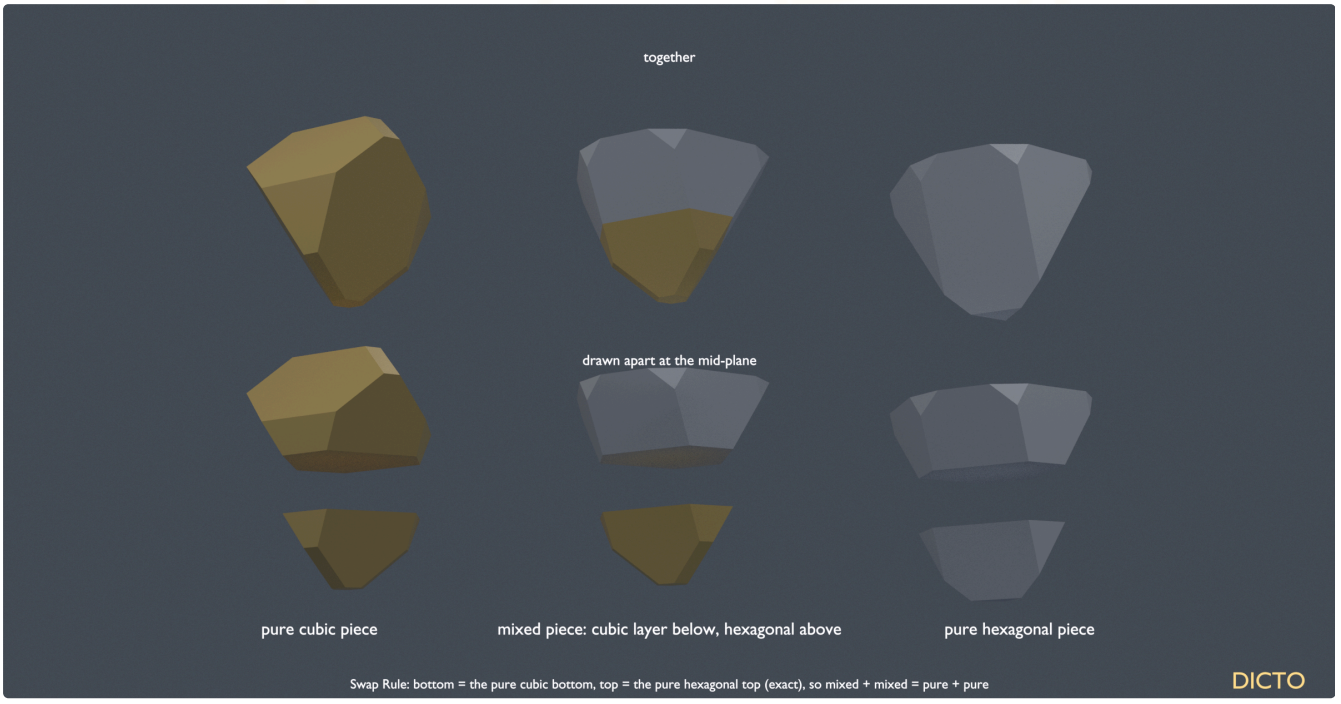


Figure 1. Ganesha. Left, the pure cubic tetrahedral piece (gold); right, the pure hexagonal one (platinum); middle, a mixed piece of the stacking ACBC (cubic layer below, hexagonal above), whole (top) and drawn apart at the mid-plane (bottom): its bottom is the cubic bottom, its top the hexagonal top. Layer spacing² = ¾·⅔, λ = 1/10; volumes of the parts 0.058005 and 0.196269.

Corollary 3 (4 : 5 : 6). At true close packing, for $5/24 < \lambda < 3/8$ the three octahedral pieces (cubic, Janus, hexagonal) have volumes in the exact ratio 4 : 5 : 6 and vanish together at $\lambda = 3/8$. Near the end of the walk each octahedral piece is cut out by its tetrahedral neighbours only. In the cubic stacking it has 8 of them, one on each face, and is a regular octahedron |x|

+ |y| + |z| ≤ 1 (volume 4/3 at that scale); in the hexagonal stacking the octahedral holes sit face to face along the stacking axis, only 6 tetrahedral neighbours remain, and the piece is the rhombohedron |x + y − z|, |x − y + z|, |−x + y + z| ≤ 1 (volume 2). Hence 4 : 6, and the Janus piece, half of each, gives 5.

4. The map

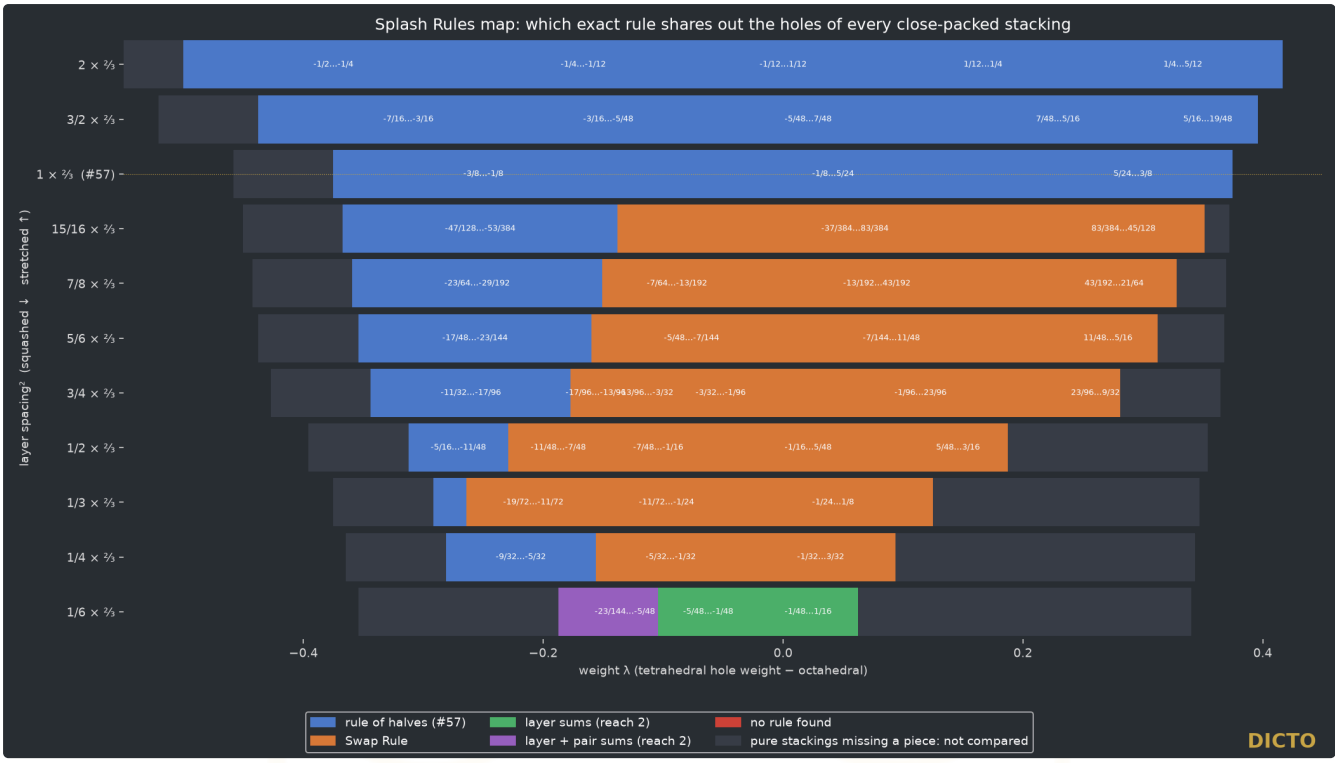


Figure 2. Which exact rule shares out the holes, for eleven layer spacings (rows; squashed below, stretched above; #57 dotted) and every window of the weight walk (columns). Blue: rule of halves only. Orange: Swap Rule with Ganesha pieces. Green and purple (s = 1/6): pieces reach two gaps; each volume is a sum of one share per layer reached (green), plus one share per neighbouring pair of layers (purple). Grey: a pure stacking lacks the piece, nothing to compare. Every window has a rule. Stackings with up to six layers; all values exact.

The map was made by the rule finder: each piece is labelled by the layer pattern around it, the shortest label that fixes its exact volume is found, and each volume is fitted exactly as a combination of the pure ones. At every spacing tested from 15/16 down to ¼ the Swap Rule appears, so it switches on as soon as the layers come closer than true close packing; the theorem proves this for s ≥ ½.

5. Proof

One polyhedron for everything. Use lateral coordinates x, y, the height ζ in layer spacings and q = d² = 2s/3, and put u = q·ζ. Then every power-distance comparison between two holes becomes a linear inequality in the five variables (q, λ, x, y, u). So a piece, its cut at the mid-plane (u ≤ q/2) and the whole range of spacings and weights (q ≥ ⅓, λ₋ ≤ λ ≤ U, all linear in q) form one convex polyhedron in five dimensions.

Locality, for all spacings at once. The lower half of a piece is fixed by the holes of its own gap and of the gap below, which depend only on the type t of the lower layer. That no other hole of any stacking cuts into it is a linear statement on the polyhedron, proved by one exact Farkas certificate per outside hole: 1070 certificates, all checked in exact rational arithmetic, covering every s ≥ ½ (s → ∞ included) and every λ in range. Lateral and vertical domination

arguments reduce the outside holes to finite lists. The upper half follows by the mirror $\zeta \to 1 - \zeta$.

Range of the walk. Summing bisector inequalities against neighbours whose weighted offsets cancel shows when a piece must be empty; this gives the bounds λ_- and U (four cases, symbolic in q and λ).

Border. For $q \geq \frac{2}{3}$ ($s \geq 1$), 38 certificates show that the holes of the gap below never reach a tetrahedral piece's far half, so it is the same for c and h. For $\frac{1}{2} \leq s < 1$, six holes present only below a c layer cut the h far half (the cutting function is 1/12 or 1/6 at $s = \frac{1}{2}$, 0 at $s = 1$, and concave), twelve holes present only below an h layer do not cut the c far half, and interior points are exhibited: the c far half is a proper part of the h far half.

Sharpness. At $s = \frac{1}{2}$ exactly, the octahedral hole of the next gap first reaches past the mid-plane (the cutting function is 0 there and positive just below). At $s = 2/5$ the exact engine finds halves that are not the pure ones, and above $U(s)$ ($s = \frac{3}{4}$, $\lambda = 5/16$) the halves rule and the Swap Rule both fail. As an independent check, the engine of #57, rebuilding whole stackings from scratch, agrees with the theorem in 4992 of 4992 exact comparisons (six stackings, $s = \frac{1}{2}$, 5/8, $\frac{3}{4}$, 1, 3/2, 2).

6. Below half the squash

Below $s = \frac{1}{2}$ the halves are no longer pure, yet the Swap Rule for volumes still holds.

> **Theorem 2.** For $\frac{1}{4} \leq s < \frac{1}{2}$ and every λ up to $3s/8$, in every close-packed stacking, each hole piece has volume $V(t|t') = A(t) + B(t')$: one term fixed by the lower layer's type, one by the upper layer's. Hence the Swap Rule $V(c|h) + V(h|c) = V(c|c) + V(h|h)$ holds for every $s \geq \frac{1}{4}$. The bound $\frac{1}{4}$ is exact.

Why. Below $s = \frac{1}{2}$ the octahedral hole of the next gap reaches across the mid-plane and carves a fixed piece D out of the lower part of this gap's octahedral and tetrahedral pieces, but only when the layer it shares is cubic. D lies against the mid-plane, out of reach of every hole of the gap below that differs between a cubic and a hexagonal layer, so it is the same piece whatever the far layer is. The lower part of a piece is therefore the pure lower half minus D when the upper layer is c; by the mirror, the upper part is the pure upper half minus a second fixed piece when the lower layer is c. Each correction is one constant per cubic layer, so the volume stays a sum of one share per layer. The halves fail only because D sits on one side of the mid-plane: in the pure cubic stacking every piece loses it, in a mixed one only some do. In exact numbers, at $s = 2/5$ and $\lambda = -29/100$ the octahedral pieces move $\delta = 2/3125$ across the mid-plane inside the same piece, and the two mixed tetrahedral pieces carry $+7/37500$ and $-7/37500$, which cancel in the sum.

Proof. The same five-variable polyhedron as in §5, now for q from 1/6 to $\frac{1}{3}$: 1706 exact Farkas certificates show that no other hole cuts a lower part and that, on each carved region, every hole that differs between c and h is inactive. At $s = \frac{1}{4}$ exactly, every hole two gaps away stops cutting (13 critical cases; the cutting function is 0 at $q = 1/6$ and changes sign there). At $s = 1/5$ the exact engine shows octahedral volumes depending on all four layers round the gap and tetrahedral ones on three, so the two-type rule fails. At $s = 1/6$ the pieces reach two gaps and the layer-sum rules of Figure 2 take over.

7. Prior art and credit

Power diagrams are affine in the weights, and absorbing the layer spacing into the weights ($u = q \cdot \zeta$) is that standard fact used once more; Farkas certificates are textbook linear programming. The rule of halves and the Janus c/h pieces Chucky and Fang are DICTO's #57. No source checked states a halves rule or a Swap Rule for hole pieces, or treats the layer spacing as a parameter of the hole power diagram.

> **Sources checked** (27 October 2026; full record with queries: study folder, prior-art.md and proof/prior-art.md). *Library* (1007 references, semantic search, 10 queries): Leppmeier 2019, the dodecahedron as a Voronoi cell; Blatov, Voronoi–Dirichlet polyhedra in crystal chemistry; Garber, Voronoi's conjecture; Hales 1998, overview of the Kepler conjecture; organizing principles for dense packings (arXiv 1206.3716); Hart, Hansen, Kuhs 2021, Markov description of stacking-disordered crystals; Pauling 1929; Torquato and Jiao 2009; Schymura, Seidel and Weltge 2021, lifts for Voronoi cells; Chaidee and Sugihara 2017, spherical Laguerre diagrams; Fedorov 1885, ch. 13. *Web*: IUCr teaching pamphlet 5 on close packing; International Tables Vol. C ch. 9.2 on layer stacking; Bourne, Kok, Roper and Spanjer, Laguerre tessellations and polycrystals (arXiv 1912.07188); Chaidee and Sugihara (arXiv 1705.03911); Voronoi volumes and interstitials in metals (arXiv 2301.04303); VoroTop defect identification. None states the rule.

8. Conclusion

True close packing is a border. At or above it, the holes of any stacking are shared out by the rule of halves of #57, and the tetrahedral pieces are whole. Below it, down to half the squash, every hole piece is a top from one pure stacking and a bottom from the other; the mixed tetrahedra become Ganesha pieces, and the two mixed pieces always add up to the two pure ones. Further down, to a quarter of the squash, the halves give way but the sums do not. However the layers are stacked, the space is still built from the halves of two pure orders.

> **Version 2** (27 October 2026): §6 now proves the Swap Rule for volumes down to $s = ¼$ (Theorem 2), which version 1 left open.

> **Data and code.** Study 2026-10-27-splash-rules: rule finder (splash.py), map (render_map.py, data/s_*.json), Ganesha check (swap_cut.py, 224/224), proof scripts and logs (proof/; PROOF.md Parts I and II list what each proves; additive.py, additive_proof.py). References: DICTO, The Rule of Halves (#57), DICTO DOI 10.5281/zenodo.23647898; DICTO, DICTO Janus c/h Pyrochlore (#60), DICTO DOI 10.5281/zenodo.23678750.